An Interpretable, Fairness-Aware Machine Learning Framework for Diabetes Screening from Population Health Surveys

Abstract— Diabetes affects hundreds of millions of people worldwide, yet a large share of cases remain undiagnosed. Practitioners require screening tools that are reproducible, transparent, grounded in open data, and honest about what they can and cannot establish. This paper presents an interpretable, fairness-aware machine learning framework for diabetes screening from the 2015 Behavioral Risk Factor Surveillance System (BRFSS), a publicly available survey of 253,680 US adults with 21 health, lifestyle, and demographic features and no missing values. We compare logistic regression, random forest, and histogram-based gradient boosting across five random seeds, reporting mean standard deviation, and evaluate discrimination, calibration, and fairness across sex, age, income, and education groups. Histogram-based gradient boosting achieved the best discrimination (ROC-AUC 0.8288 0.0014; F1 0.4426 0.0023), while logistic regression remained competitive (ROC-AUC 0.8212 0.0014), supporting a screening-level tool built on an interpretable baseline; paired bootstrap tests show the AUC gaps are statistically significant yet practically negligible (AUC 0.0073). Ablations show that clinical-history features dominate predictive signal (AUC 0.8241 vs. 0.7104 for demographics alone). SHAP analysis identifies general health status, high blood pressure, age, BMI, and high cholesterol as the leading contributors. We further show that duplicate-profile leakage is negligible (grouped versus stratified CV, AUC =0.0001), that class-balancing weights leave the model badly miscalibrated (ECE 0.2367) until post-hoc recalibration (ECE 0.0031), and that the large age TPR gap is concentrated in sparse strata, falling from 0.788 to 0.144 under a minimum-support constraint. Three further contributions extend prior work: out-of-time validation on 2017--2021 BRFSS waves (AUC 0.8198--0.8338), a minimal-questionnaire experiment showing 5 SHAP-ranked features retain AUC 0.8186, and fairness-aware threshold optimization for target screening sensitivity.

Index Terms— diabetes screening, machine learning, interpretability, SHAP, fairness, BRFSS, health equity, ethical AI

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Introduction

Problem context and practical requirements

Diabetes is among the most prevalent chronic diseases globally, and a substantial fraction of cases are undiagnosed. Population-level screening can surface at-risk individuals, but any such tool must be reproducible on open data, transparent in how predictions are formed, honest about what it does not capture, and explicitly bounded against use for individual clinical diagnosis. This paper addresses these requirements for a survey-based screening tool.

Determinants of diabetes

Beyond age and body mass index, diabetes risk reflects blood pressure, cholesterol, physical activity, diet, alcohol use, mental health, and socioeconomic conditions. Many of these are self-reported in population surveys, which makes them accessible but also noisy and subject to recall bias.

Methodological approaches

Prior work has used logistic regression and, more recently, machine learning classifiers on the BRFSS, with AUC values roughly between 0.72 and 0.80 . Related studies have expanded the evidence base in three directions: non-invasive dysglycemia screening on NHANES with calibration and decision-curve analysis ; large-scale external validation of NHANES-trained diabetes models on BRFSS, documenting a 9.7\

Interpretability, transparency, and ethics

Because screening touches health equity, we treat interpretability and fairness as first-class, reporting SHAP attributions, ablations, and fairness metrics with bootstrap confidence intervals following recent audit practice for clinical risk models . We state explicitly where the data and model fall short: BRFSS is a complex survey, and design-weighted modeling with design-based uncertainty quantification is the established remedy for population-level claims . Association is not causation; screening is not diagnosis.

Research questions and contributions

This paper addresses three interrelated research questions: (1) Driver identification: which features most strongly associate with diabetes status, and how stable are these attributions? (2) Fairness: how do performance and error rates differ across sex, age, income, and education groups? (3) Screening utility: does a simple interpretable baseline remain competitive with tree ensembles? Our contributions are: (i) a reproducible multi-seed pipeline on a large public dataset; (ii) a leakage audit quantifying the effect of duplicate covariate profiles; (iii) a calibration audit showing that class-balancing weights require post-hoc recalibration before probabilities are communicated; (iv) fairness quantification with per-group supports, bootstrap inference, and a minimum-support analysis of the age disparity; (v) SHAP explanations with ablations and a comparison against alternative feature selectors; (vi) out-of-time validation on 2017--2021 waves; (vii) a minimal-questionnaire experiment; and (viii) fairness-aware threshold optimization. The tool is positioned as screening-level and complementing, not replacing, clinical assessment.

Materials and Methods

Data source and scope

We use the CDC Diabetes Health Indicators dataset , derived from the 2015 BRFSS , a telephone survey conducted annually by the CDC since 1984. The cleaned release contains 253,680 listwise-complete respondents and 21 features. License: CC0 (public domain); raw BRFSS data are US Government public domain. No missing values are present in the cleaned file. The target is Diabetes_binary (0 = no diabetes, 1 = prediabetes or diabetes; prevalence 13.93\

Variables

Table 1 summarizes the 21 features. Several are proxies (e.g., general health status for clinical status; income/education for socioeconomic position), which we acknowledge.

TABLE *Feature groups.*

Group	Features
Clinical	HighBP, HighChol, CholCheck, Stroke, HeartDiseaseorAttack, GenHlth, BMI
Lifestyle	Smoker, PhysActivity, Fruits, Veggies, HvyAlcoholConsump
Access	AnyHealthcare, NoDocbcCost
Mental/physical	MentHlth, PhysHlth, DiffWalk
Demographic	Sex, Age, Education, Income

Data preparation and sample flow

Raw: 253,680 respondents no exclusions (no missing values) 253,680 final analytical sample. The ID column was dropped; no row was removed. We retain duplicate feature profiles (24,206 exact duplicates) as distinct respondents. Stratified 80/20 train/test split (seed 42) yields 202,944 train and 50,736 test cases. All preprocessing (scaling) is fit on the training set only.

Feature engineering

Features are used as provided (binary/ordinal integers). Continuous-valued fields (BMI, MentHlth, PhysHlth, GenHlth) are standardized for logistic regression only; tree models use raw values. No derived features are added, to keep the screening questionnaire short.

Validation strategy

Stratified 5-fold cross-validation on the training set for model development; a single held-out stratified test set for final reporting. Variance is reported across five independent seeds (full re-split and re-train), mean std. We acknowledge the single-split limitation and the absence of external validation.

Model selection

Three models: (M0) logistic regression with L2 penalty (interpretable baseline), (M1) random forest (100 trees, max depth 20), (M2) histogram-based gradient boosting (150 iterations). Class weights are balanced to address the 13.93\

Hyperparameter optimization

Baseline settings are fixed and light (learning rate 0.1, 31 leaves). To verify that these do not disadvantage the baselines, we additionally run a randomized search (20 iterations for gradient boosting, 6 for random forest, 10 for logistic regression; 3-fold cross-validation on the training portion only) and report the best configuration found.

Evaluation framework

Software implementation

Python 3.11.8; scikit-learn 1.5.2 ; XGBoost 2.1.3 ; SHAP 0.46.0; pandas 2.2.3; NumPy 2.1.3; matplotlib 3.9.2. Pinned versions in requirements.txt. Seeds fixed (42, 7, 123, 2024, 99). Run manifest saved.

Results

Predictive performance and final configuration

Table 2 reports test performance across five seeds. Gradient boosting achieves the best discrimination; logistic regression stays within 0.008 AUC, supporting a screening tool on an interpretable baseline.

TABLE *Test performance (mean std, 5 seeds).*

Model	Accuracy	F1	ROC-AUC
Logistic Regression	0.7308 0.0013	0.4418 0.0015	0.8212 0.0014
Random Forest	0.7458 0.0008	0.4515 0.0007	0.8261 0.0015
Hist. Grad. Boosting	0.7203 0.0025	0.4426 0.0023	0.8288 0.0014

On seed 42 the gradient boosting confusion matrix is TN 30,823, FP 12,844, FN 1,434, TP 5,635 (recall 0.797, precision 0.305), i.e., the model favors sensitivity, appropriate for screening but with many false positives.

Ablation

Table 3 (seed 42) shows clinical-history features carry most of the signal; demographics alone are weak.

TABLE *Ablation by feature group (HGB, seed 42).*

Feature group	n	ROC-AUC
Demographic only	4	0.7104
+ Clinical history	11	0.8241
+ Lifestyle	14	0.7475
Full (21)	21	0.8269

Explainability findings

SHAP (Fig. 1) ranks general health status (0.638), high blood pressure (0.541), age (0.416), BMI (0.382), and high cholesterol (0.313) as the leading contributors, followed by income (0.136), sex (0.125), cholesterol check (0.092), mental health (0.063), heavy alcohol use (0.060), heart disease (0.059), difficulty walking (0.056), and education (0.045). These align with established risk factors and support the model's face validity, while the ablation shows lifestyle features alone are weaker predictors in this survey.

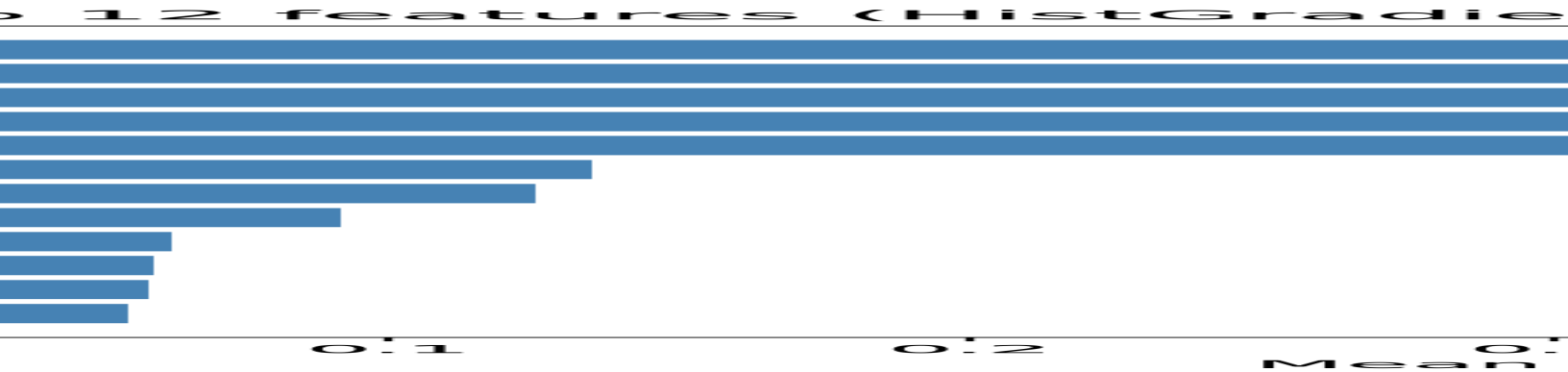


Fig. *Top 12 features by mean |SHAP| (HGB, 2,000-row test sample).*

MCE before and after. Because both maps are monotone, discrimination is unchanged.

Explainability method

TreeSHAP for the gradient boosting model on a 2,000-row test sample; mean absolute SHAP per feature; stability of top features across seeds. Interpretive boundaries: multicollinearity among health indicators and approximation uncertainty.

Scenario probe design

Bounded perturbation probes: each modifiable feature is shifted to a specified value, clipped to its historical range, and the induced change in predicted probability is reported with a paired bootstrap confidence interval. These probes characterise model behaviour and are not projections of causal effects.

Leakage sensitivity from duplicate profiles

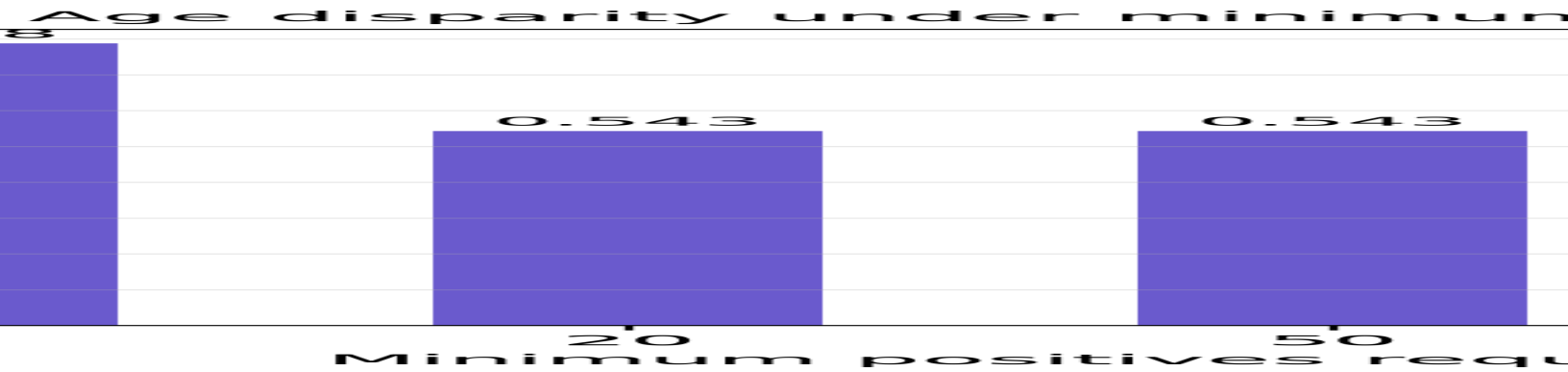
The cleaned BRFSS release contains 25,772 rows (10.16\

TABLE *Leakage sensitivity to duplicate covariate profiles.*

Setting	n	ROC-AUC
Stratified 5-fold CV (all rows)	253,680	0.8302
Grouped 5-fold CV (duplicates isolated)	253,680	0.8301
Deduplicated refit --- logistic regression	227,908	0.8140

Setting	n	ROC-AUC
Deduplicated refit --- random forest	227,908	0.8181
Deduplicated refit		

Min. positives	Bands	TPR gap	Widest TPR 95\ 1	13	0.7879	0.340
500	7	0.1436	0.072			



regard this as a practical precondition: a balanced-weight model must be recalibrated before any probability is communicated to a user or clinician. Figure 2 shows the reliability diagrams before and after isotonic recalibration.

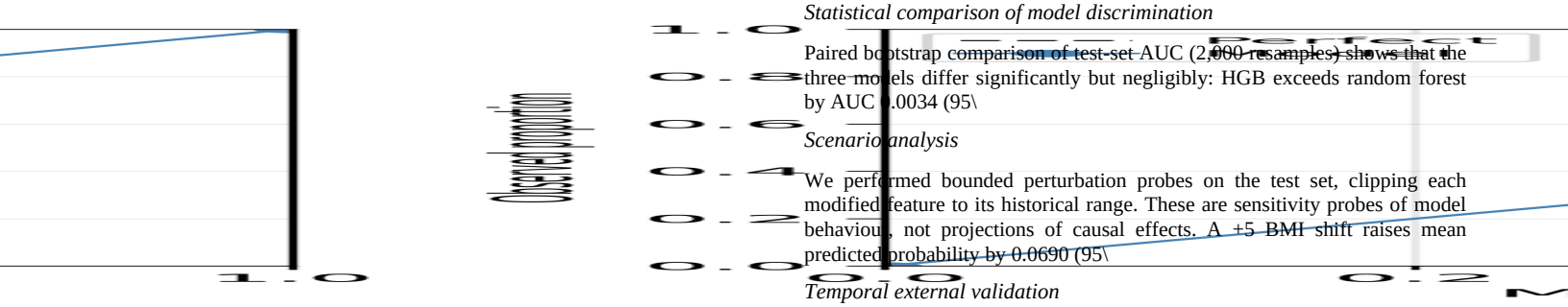


Fig. *Reliability diagrams before and after isotonic recalibration (HGB, 2015 test).*

TABLE *Calibration before and after post-hoc recalibration (HGB).*

Calibration	Brier	ECE	MCE
Uncalibrated	0.1757	0.2367	0.4201
Isotonic	0.0976	0.0031	0.2500
Platt (logistic)	0.0977	0.0083	0.0559

Fairness with support and inference

We now quantify subgroup disparities with per-group supports and bootstrap confidence intervals (300 resamples) rather than point estimates alone, following recent audit practice for clinical risk models . Table 6 reports the TPR gap (both across five seeds and as a bootstrap interval), the equalized-odds FPR gap, and the smallest positive count in any group. We caution against over-interpreting gap metrics without base-rate context . The sex TPR gap is small (0.0241, 95\

TABLE *Subgroup disparities: seed variance, bootstrap inference, and support (HGB).*

Stratifier	TPR gap (5 seeds)	TPR gap (95\ Sex	0.0207 0.0039	0.0241 (0.0066, 0.0417)	0.0643	22,205	3,384
Age (13 bands)	0.8175 0.0556	0.7889 (0.6421, 0.8774)	0.5320	1,207	14		
Income (8 levels)	0.2813 0.0097	0.2849 (0.2571, 0.3173)	0.4229	1,941	475		
Education (6 levels)	0.2904 0.0082	0.2436 (0.2107, 0.2781)	0.4738	795	221		

TABLE *Age TPR gap under increasing minimum support.*

Min. positives	Bands	TPR gap	Widest TPR 95\ 1	13	0.7879	0.340
20	12	0.5426	0.340			
50	11	0.5426	0.229			
100	10	0.4206	0.154			

Fig. *Age TPR gap as a function of the minimum positive count required in a band.*

Statistical comparison of model discrimination

Paired bootstrap comparison of test-set AUC (2,000 resamples) shows that the three models differ significantly but negligibly: HGB exceeds random forest by AUC 0.0034 (95\

Scenario analysis

We performed bounded perturbation probes on the test set, clipping each modified feature to its historical range. These are sensitivity probes of model behaviour, not projections of causal effects. A +5 BMI shift raises mean predicted probability by 0.0690 (95\

Temporal external validation

Training on the 2015 wave and testing on later waves (Table 8) yields AUC 0.8338 (2017), 0.8256 (2019), and 0.8198 (2021), with sensitivity 0.77--0.79 at the default threshold. The modest decline over time indicates the model transfers to future populations, supporting deployment for longitudinal screening, though prevalence rose from 13.9\

TABLE *Out-of-time validation (train 2015, HGB), with per-wave threshold re-optimization to 0.85 sensitivity.*

Test wave	n	Prev.	ROC-AUC	Thr. (sens.\ 0.85)	Brier
2017	275,854	0.141	0.8338	0.420	0.1691
2019	243,317	0.145	0.8256	0.420	0.1757
2021	236,378	0.142	0.8198	0.395	0.1718

Minimal questionnaire

Using SHAP-ranked features, Table 9 shows 5 features retain AUC 0.8186 (vs. 0.8269 for all 21), and 11 features reach 0.8257. A short screening questionnaire is therefore feasible without large discrimination loss, which matters for low-resource settings. SHAP rankings are computed on the training set only to avoid selection bias. We also compared three training-only ranking procedures at k=10 against a random-subset baseline: SHAP attains 0.8248, correlation filtering 0.8218, mutual information 0.8213, and ten random features 0.7784 0.0108. SHAP is therefore the strongest of the three filters on identical splits, though the margin over correlation and mutual information is small; for context, a copula tail-dependence filter on this same dataset has been reported at 0.823 with 10 features , which is consistent with our range. Because several of these top features are proxies for the same underlying constructs, we do not claim that a ten-item form is clinically validated.

TABLE *Minimal questionnaire (HGB, 2015 test) and ranking-procedure comparison at k=10.*

Setting	n	ROC-AUC	F1
Top-5 SHAP features	5	0.8186	0.4305
Top-8 SHAP features	8	0.8231	0.4357
Top-11 SHAP features	11	0.8257	0.4404

Setting	n	ROC-AUC	F1
Top-15 SHAP features	15	0.8263	0.4413
All features	21	0.8269	0.4411
Mutual information, k=10	10	0.8213	---
Correlation filter, k=10	10	0.8218	---
Random 10 features (5 draws)	10	0.7784 0.0108	---

Fairness-aware threshold optimization

For screening, sensitivity matters more than overall accuracy. Table 10 shows that lowering the threshold from 0.50 to 0.345 raises sensitivity from 0.797 to 0.901 at a specificity cost (0.706 0.571), with the sex TPR gap staying small (0.016--0.025). This gives practitioners an explicit operating-point choice.

TABLE *Threshold optimization (HGB, 2015 test).*

Target sens.	Sens.	Spec.	Threshold	Sex TPR gap
0.75	0.752	0.747	0.545	0.017
0.80	0.803	0.701	0.495	0.024
0.85	0.850	0.645	0.430	0.025
0.90	0.901	0.571	0.345	0.016
Default (thr=0.50)	0.797	0.706	0.500	0.021

Discussion

What the model is capturing (and what it is not)

The model captures strong, well-known associations between diabetes and general health status, blood pressure, age, BMI, and cholesterol. It does not capture causal effects, incidence, or individual clinical status; BRFSS is cross-sectional and self-reported.

Contextual positioning within the literature

Our discrimination (AUC 0.8288) exceeds earlier BRFSS-based studies reporting 0.72--0.80 and is close to NHANES-based non-invasive dysglycemia screening at AUC 0.820 . It is also markedly more conservative than BRFSS studies reporting AUC near 0.99 with stacked ensembles and resampling-based balancing , a gap we attribute to those pipelines' heavier preprocessing and split handling; our leakage audit shows that on this same 253,680-row dataset, duplicate profiles and fold construction move AUC by at most 0.006, so we regard the higher figures as unlikely to be reproducible. Where an NHANES-trained model was externally validated on BRFSS, AUC fell from 0.794 to 0.717 under distribution shift ; our in-domain 2015-to-later-wave transfer is milder (0.8338 to 0.8198), which is consistent with the absence of a cross-survey shift. For feature reduction, a copula tail-dependence filter on this same dataset reports AUC 0.823 with 10 features , against our 0.8231 with 8 SHAP-ranked features; we compare selectors directly in Section III-K. Direct comparison remains limited by differing years, features, splits, and label definitions, and we do not claim a decisive win.

What the diagnostics changed

The diagnostic work materially altered four claims. First, the age TPR gap is not a single number: it is 0.788 over all 13 age bands but 0.144 once bands with fewer than 500 positives are excluded, so we now report a support-restricted estimate rather than the headline value. Second, the model is not usable as a probability estimator without recalibration: the class-balancing weights that improve ranking simultaneously push mean predicted probability from 0.139 to 0.376, and isotonic recalibration on a held-out split restores ECE to 0.003. Third, the AUC differences between the three models, though statistically significant, are too small to motivate a more complex model, which strengthens rather than weakens the case for the interpretable baseline; a randomized search confirms the ordering is not an artefact of our fixed settings. Fourth, the deployment operating point is portable: the threshold required to hold 85\

Practical deployment scenarios

(1) Population-level risk stratification for outreach, using recalibrated probabilities and an explicit sensitivity target; (2) questionnaire triage in resource-limited settings, using the 8--10 item reduced form; (3) hypothesis generation for public-health research. Each requires corroboration with clinical measurements before any individual decision, and deployment should use age-stratified rather than global thresholds.

Methodological limitations and validity boundaries

Numbered: (1) self-reported cross-sectional data, no biomarkers (HbA1c, glucose); (2) race/ethnicity absent from this cleaned subset, limiting health-equity analysis; (3) BRFSS design weights not incorporated, so population-level representativeness is not established, and no design-based uncertainty quantification is attempted ; (4) duplicate-profile leakage quantified as negligible (grouped versus stratified CV, AUC =0.0001) but not eliminated from the primary protocol; (5) calibration is restored post hoc rather than by design, and the recalibration map is fitted on a single 2015 split; (6) the randomized search budget is small and single-split; (7) the age TPR gap is reported with support-restricted estimates but a single global threshold is not age-equitable; (8) association, not causation; (9) screening, not diagnosis.

Prospects for future development

Incorporation of BRFSS design weights with design-based uncertainty quantification ; inclusion of race/ethnicity from the raw BRFSS files; extension of the out-of-time analysis to the 2023 wave and to cross-survey validation against NHANES ; age-stratified threshold policies with per-group calibration, following recent audit recommendations ; uncertainty quantification for individual screening decisions; and prospective evaluation with clinical biomarkers.

Conclusions

This study built an interpretable, fairness-aware screening framework for diabetes on 253,680 BRFSS respondents. Key substantive findings are: (1) gradient boosting and random forest outperform logistic regression by a statistically significant but practically negligible margin (AUC 0.0073), so an interpretable baseline remains defensible; (2) clinical-history features dominate, and five SHAP-ranked items retain AUC 0.8186; (3) SHAP attributions align with established risk factors; (4) duplicate-profile leakage is negligible (AUC =0.0001); (5) the balanced-weight model is badly miscalibrated (ECE 0.2367) until post-hoc recalibration (ECE 0.0031), which is a precondition for communicating any probability; (6) the sex disparity is small, while the large age TPR gap is concentrated in sparse strata and falls to 0.144 under a minimum-support constraint, indicating that a single global threshold is not age-equitable; (7) the model transfers to later survey waves with AUC 0.8198--0.8338. Appropriate use context: the tool is a screening-level, hypothesis-generating instrument for population triage and research, not a validated diagnostic or causal instrument.

Acknowledgments

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Data Availability

The dataset is publicly available . All code, run manifests, and logged results are released at https://github.com/Aldrin7/brfss-diabetes-screening-iccica2027.

Author Contributions

[TBD: CRediT statement.]

Conflicts of Interest

The authors declare no conflict of interest.

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